

Quantum Mechanics and Solid State Physics

Numerical Exercises with Step-by-Step Solutions

1 Hydrogen Atom Energy Levels

The energy levels of the hydrogen atom are given by:

$$E_n = -\frac{13.6 \text{ eV}}{n^2} \quad (1)$$

where $n = 1, 2, 3, \dots$ is the principal quantum number.

Exercise 1.1: Energy of Ground State and First Excited State

Problem: Calculate the energy of the ground state ($n = 1$) and the first excited state ($n = 2$) of the hydrogen atom. What is the energy difference between these two states?

Solution

Step 1: Identify the quantum numbers

For the ground state: $n = 1$

For the first excited state: $n = 2$

Step 2: Calculate the ground state energy

Using the formula $E_n = -\frac{13.6 \text{ eV}}{n^2}$:

$$E_1 = -\frac{13.6 \text{ eV}}{1^2} = -13.6 \text{ eV} \quad (2)$$

The negative sign indicates that the electron is bound to the nucleus.

Step 3: Calculate the first excited state energy

$$E_2 = -\frac{13.6 \text{ eV}}{2^2} = -\frac{13.6 \text{ eV}}{4} = -3.4 \text{ eV} \quad (3)$$

Step 4: Calculate the energy difference

The energy difference is:

$$\Delta E = E_2 - E_1 = -3.4 \text{ eV} - (-13.6 \text{ eV}) = 10.2 \text{ eV} \quad (4)$$

Physical Interpretation:

An electron transitioning from $n = 2$ to $n = 1$ would emit a photon with energy 10.2 eV. This is in the ultraviolet region of the electromagnetic spectrum and corresponds to the Lyman-alpha line.

Answer: $E_1 = -13.6 \text{ eV}$, $E_2 = -3.4 \text{ eV}$, $\Delta E = 10.2 \text{ eV}$

Exercise 1.2: Wavelength of Emitted Photon

Problem: An electron in a hydrogen atom transitions from the $n = 3$ state to the $n = 2$ state. Calculate the wavelength of the emitted photon. (Use $hc = 1240 \text{ eV}\cdot\text{nm}$)

Solution

Step 1: Calculate the energy of the $n = 3$ state

$$E_3 = -\frac{13.6 \text{ eV}}{3^2} = -\frac{13.6 \text{ eV}}{9} = -1.511 \text{ eV} \quad (5)$$

Step 2: Calculate the energy of the $n = 2$ state

$$E_2 = -\frac{13.6 \text{ eV}}{2^2} = -\frac{13.6 \text{ eV}}{4} = -3.4 \text{ eV} \quad (6)$$

Step 3: Calculate the energy of the emitted photon

The energy of the photon equals the energy difference between the two levels:

$$E_{\text{photon}} = E_3 - E_2 = -1.511 \text{ eV} - (-3.4 \text{ eV}) = 1.889 \text{ eV} \quad (7)$$

Step 4: Calculate the wavelength using $E = \frac{hc}{\lambda}$

Rearranging for wavelength:

$$\lambda = \frac{hc}{E_{\text{photon}}} = \frac{1240 \text{ eV} \cdot \text{nm}}{1.889 \text{ eV}} = 656.4 \text{ nm} \quad (8)$$

Physical Interpretation:

This wavelength is in the visible red region of the spectrum and corresponds to the H-alpha line in the Balmer series, which is commonly observed in astronomical spectra and gives hydrogen emission nebulae their characteristic red color.

Answer: $\lambda = 656.4 \text{ nm}$ (red visible light)

Exercise 1.3: Ionization Energy from Excited State

Problem: How much energy is required to ionize a hydrogen atom that is initially in the $n = 3$ state? Compare this to the ionization energy from the ground state.

Solution

Step 1: Understand ionization

Ionization means removing the electron completely from the atom. A free electron has zero energy ($E_\infty = 0$), so we need to provide enough energy to bring the electron from its initial state to $E = 0$.

Step 2: Calculate the energy of the $n = 3$ state

$$E_3 = -\frac{13.6 \text{ eV}}{3^2} = -\frac{13.6 \text{ eV}}{9} = -1.511 \text{ eV} \quad (9)$$

Step 3: Calculate ionization energy from $n = 3$

The ionization energy is the energy needed to go from E_3 to $E_\infty = 0$:

$$E_{\text{ion}}(n = 3) = E_\infty - E_3 = 0 - (-1.511 \text{ eV}) = 1.511 \text{ eV} \quad (10)$$

Step 4: Calculate ionization energy from ground state

For comparison, from the ground state:

$$E_{\text{ion}}(n = 1) = E_\infty - E_1 = 0 - (-13.6 \text{ eV}) = 13.6 \text{ eV} \quad (11)$$

Step 5: Compare the two ionization energies

$$\frac{E_{\text{ion}}(n = 3)}{E_{\text{ion}}(n = 1)} = \frac{1.511 \text{ eV}}{13.6 \text{ eV}} = \frac{1}{9} \quad (12)$$

Physical Interpretation:

An electron in the $n = 3$ state is already much less tightly bound to the nucleus than in the ground state. It requires only $1/9$ of the ground state ionization energy to remove it completely. This is consistent with the $1/n^2$ dependence of the binding energy.

Answer: Ionization energy from $n = 3$ is 1.511 eV, which is 9 times smaller than the ground state ionization energy of 13.6 eV.

2 Kronig-Penney Model for Band Structure

The Kronig-Penney model describes electrons in a periodic potential. The key result is the dispersion relation connecting energy E and wavevector k , derived from:

$$\cos(ka) = \cos(\alpha a) + \frac{P}{\alpha a} \sin(\alpha a) \quad (13)$$

where $\alpha = \sqrt{2mE}/\hbar$, a is the lattice constant, and P is a dimensionless parameter characterizing the potential strength.

Exercise 2.1: Energy at the Zone Boundary

Problem: Consider a one-dimensional Kronig-Penney model with lattice constant $a = 5 \text{ \AA}$. At the first Brillouin zone boundary ($k = \pi/a$), the dispersion relation simplifies. If $\alpha a = 2\pi$ at this point and $P = 1$, verify that the right-hand side of the Kronig-Penney equation equals $\cos(ka)$ and interpret the result.

Solution**Step 1: Identify the zone boundary condition**

At the first Brillouin zone boundary:

$$k = \frac{\pi}{a} \quad (14)$$

Therefore:

$$ka = \frac{\pi}{a} \cdot a = \pi \quad (15)$$

Step 2: Calculate the left-hand side

$$\cos(ka) = \cos(\pi) = -1 \quad (16)$$

Step 3: Calculate the right-hand side with given parameters

Given: $\alpha a = 2\pi$ and $P = 1$

$$\text{RHS} = \cos(\alpha a) + \frac{P}{\alpha a} \sin(\alpha a) \quad (17)$$

$$= \cos(2\pi) + \frac{1}{2\pi} \sin(2\pi) \quad (18)$$

$$= 1 + \frac{1}{2\pi} \cdot 0 \quad (19)$$

$$= 1 \quad (20)$$

Step 4: Compare LHS and RHS

We have $\text{LHS} = -1$ and $\text{RHS} = 1$. These are not equal!

Step 5: Physical interpretation

Since $\cos(ka)$ must lie between -1 and $+1$, and the RHS must equal the LHS for allowed energy states, this particular combination of parameters at $\alpha a = 2\pi$ does NOT represent an allowed energy state at the zone boundary.

To find allowed energies at the zone boundary, we need to solve:

$$-1 = \cos(\alpha a) + \frac{1}{\alpha a} \sin(\alpha a) \quad (21)$$

This would give a different value of αa (and thus a different energy).

Physical Interpretation:

This demonstrates that not all energies are allowed in the periodic potential—there are energy gaps (forbidden bands). The parameters given correspond to a forbidden state, illustrating the origin of band gaps in solids.

Answer: $\text{LHS} = -1$, $\text{RHS} = 1$. These are unequal, indicating this is a forbidden energy state at the zone boundary, demonstrating the existence of band gaps.

Exercise 2.2: Width of First Brillouin Zone

Problem: For a one-dimensional crystal with lattice constant $a = 4 \text{ \AA}$, calculate:

- (a) The width of the first Brillouin zone in k -space
- (b) The width in units of nm^{-1}

Solution

Step 1: Understand the first Brillouin zone

In one dimension, the first Brillouin zone extends from $k = -\pi/a$ to $k = +\pi/a$.

Step 2: Calculate the width in terms of a

The width of the first Brillouin zone is:

$$\Delta k = \frac{\pi}{a} - \left(-\frac{\pi}{a}\right) = \frac{2\pi}{a} \quad (22)$$

Step 3: Substitute the given lattice constant

Given: $a = 4 \text{ \AA} = 4 \times 10^{-10} \text{ m} = 0.4 \text{ nm}$

$$\Delta k = \frac{2\pi}{0.4 \text{ nm}} = \frac{2\pi}{0.4} \text{ nm}^{-1} = 5\pi \text{ nm}^{-1} \quad (23)$$

Step 4: Calculate numerical value

$$\Delta k = 5\pi \text{ nm}^{-1} \approx 15.71 \text{ nm}^{-1} \quad (24)$$

Physical Interpretation:

The Brillouin zone represents the range of unique k values for electron waves in the periodic lattice. Smaller lattice constants lead to wider Brillouin zones. All electronic properties of the crystal can be described using only the wavevectors within this zone due to the periodicity of the crystal lattice.

Answer:

- (a) Width $= 2\pi/a = 5\pi \text{ nm}^{-1}$
- (b) Width $\approx 15.71 \text{ nm}^{-1}$

Exercise 2.3: Free Electron Approximation

Problem: In the weak potential limit ($P \rightarrow 0$), the Kronig-Penney model reduces to the free electron case. For $P = 0$ and lattice constant $a = 5 \text{ \AA}$, calculate the energy of an electron at $k = \pi/(2a)$ using the free electron dispersion relation $E = \hbar^2 k^2/(2m)$. Express your answer in eV. (Use $\hbar^2/(2m) = 3.81 \text{ eV} \cdot \text{\AA}^2$)

Solution**Step 1: Write down the free electron dispersion relation**

For free electrons (or weak periodic potential):

$$E = \frac{\hbar^2 k^2}{2m} \quad (25)$$

Step 2: Calculate the wavevector

Given: $k = \frac{\pi}{2a}$ and $a = 5 \text{ \AA}$

$$k = \frac{\pi}{2 \times 5 \text{ \AA}} = \frac{\pi}{10} \text{ \AA}^{-1} \quad (26)$$

Step 3: Calculate k^2

$$k^2 = \left(\frac{\pi}{10}\right)^2 \text{ \AA}^{-2} = \frac{\pi^2}{100} \text{ \AA}^{-2} \quad (27)$$

Step 4: Calculate the energy

Using the given value $\frac{\hbar^2}{2m} = 3.81 \text{ eV} \cdot \text{\AA}^2$:

$$E = 3.81 \text{ eV} \cdot \text{\AA}^2 \times \frac{\pi^2}{100} \text{ \AA}^{-2} \quad (28)$$

$$E = \frac{3.81\pi^2}{100} \text{ eV} = \frac{3.81 \times 9.8696}{100} \text{ eV} \quad (29)$$

$$E = \frac{37.60}{100} \text{ eV} = 0.376 \text{ eV} \quad (30)$$

Physical Interpretation:

This energy corresponds to a free electron with wavevector $k = \pi/(2a)$, which is halfway to the first Brillouin zone boundary. In a real crystal with non-zero potential ($P > 0$), the energy would be slightly different due to the periodic potential, and energy gaps would open at the zone boundaries.

Answer: $E = 0.376 \text{ eV}$

3 Effective Mass Calculation for Electrons and Holes

The effective mass describes how electrons and holes respond to external forces in a crystal. It is defined by:

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2} \quad (31)$$

where $E(k)$ is the energy dispersion relation.

Exercise 3.1: Effective Mass from Parabolic Dispersion

Problem: Near the bottom of a conduction band, the dispersion relation is $E(k) = E_c + \frac{\hbar^2 k^2}{2m_e^*}$, where $E_c = 1.5$ eV and $m_e^* = 0.067m_0$ (with m_0 being the free electron mass). Calculate:

- (a) The effective mass in units of m_0
- (b) The energy of an electron at $k = 0.1 \text{ \AA}^{-1}$ relative to the band edge

(Use $\hbar^2/(2m_0) = 3.81 \text{ eV} \cdot \text{\AA}^2$)

Solution**Step 1: Calculate the second derivative of $E(k)$**

Given: $E(k) = E_c + \frac{\hbar^2 k^2}{2m_e^*}$

Taking the first derivative:

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m_e^*} \quad (32)$$

Taking the second derivative:

$$\frac{d^2 E}{dk^2} = \frac{\hbar^2}{m_e^*} \quad (33)$$

Step 2: Apply the effective mass definition

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2} = \frac{1}{\hbar^2} \cdot \frac{\hbar^2}{m_e^*} = \frac{1}{m_e^*} \quad (34)$$

Therefore: $m^* = m_e^* = 0.067m_0$

Answer to part (a): The effective mass is $m^* = 0.067m_0$

Step 3: Calculate the energy at $k = 0.1 \text{ \AA}^{-1}$

The energy relative to the band edge (E_c) is:

$$\Delta E = E(k) - E_c = \frac{\hbar^2 k^2}{2m_e^*} \quad (35)$$

We can rewrite this using:

$$\frac{\hbar^2}{2m_e^*} = \frac{\hbar^2}{2m_0} \cdot \frac{m_0}{m_e^*} = \frac{\hbar^2}{2m_0} \cdot \frac{1}{0.067} \quad (36)$$

$$\frac{\hbar^2}{2m_e^*} = 3.81 \text{ eV} \cdot \text{\AA}^2 \times \frac{1}{0.067} = 56.87 \text{ eV} \cdot \text{\AA}^2 \quad (37)$$

Step 4: Calculate ΔE

$$\Delta E = 56.87 \text{ eV} \cdot \text{\AA}^2 \times (0.1 \text{ \AA}^{-1})^2 \quad (38)$$

$$\Delta E = 56.87 \times 0.01 \text{ eV} = 0.569 \text{ eV} \quad (39)$$

Physical Interpretation:

The small effective mass ($0.067m_0$) means electrons in this material (like GaAs) respond very strongly to applied fields and have high mobility. This makes such materials excellent for high-speed electronic devices.

Answer:

(a) $m^* = 0.067m_0$

(b) $\Delta E = 0.569 \text{ eV}$ above the conduction band edge

Exercise 3.2: Heavy Hole vs. Light Hole

Problem: In many semiconductors, there are two types of holes at the valence band maximum: heavy holes with $m_{hh}^* = 0.5m_0$ and light holes with $m_{lh}^* = 0.08m_0$. For the same energy of 0.1 eV below the valence band maximum, calculate the wavevector magnitude for both types of holes. What is the ratio of their wavelengths?

Solution**Step 1: Write the dispersion relation for holes**

Near the valence band maximum at E_v , holes have energy:

$$E = E_v - \frac{\hbar^2 k^2}{2m_h^*} \quad (40)$$

The energy below the valence band edge is:

$$|E - E_v| = \frac{\hbar^2 k^2}{2m_h^*} \quad (41)$$

Step 2: Solve for k

$$k = \sqrt{\frac{2m_h^* |E - E_v|}{\hbar^2}} \quad (42)$$

We can express this as:

$$k = \sqrt{\frac{m_h^*}{m_0}} \times \sqrt{\frac{2m_0 \times 0.1 \text{ eV}}{\hbar^2}} \quad (43)$$

Step 3: Calculate the common factor

Using $\frac{\hbar^2}{2m_0} = 3.81 \text{ eV} \cdot \text{\AA}^2$:

$$\frac{2m_0}{\hbar^2} = \frac{1}{3.81} \text{ eV}^{-1} \cdot \text{\AA}^{-2} \quad (44)$$

$$\sqrt{\frac{2m_0 \times 0.1 \text{ eV}}{\hbar^2}} = \sqrt{\frac{0.1}{3.81}} \text{\AA}^{-1} = 0.162 \text{\AA}^{-1} \quad (45)$$

Step 4: Calculate k for heavy holes

$$k_{hh} = \sqrt{0.5} \times 0.162 \text{\AA}^{-1} = 0.115 \text{\AA}^{-1} \quad (46)$$

Step 5: Calculate k for light holes

$$k_{lh} = \sqrt{0.08} \times 0.162 \text{\AA}^{-1} = 0.0458 \text{\AA}^{-1} \quad (47)$$

Step 6: Calculate wavelengths

The wavelength is $\lambda = 2\pi/k$:

$$\lambda_{hh} = \frac{2\pi}{0.115 \text{\AA}^{-1}} = 54.6 \text{\AA} \quad (48)$$

$$\lambda_{lh} = \frac{2\pi}{0.0458 \text{\AA}^{-1}} = 137.2 \text{\AA} \quad (49)$$

Step 7: Calculate the wavelength ratio

$$\frac{\lambda_{lh}}{\lambda_{hh}} = \frac{137.2}{54.6} = 2.51 \approx \sqrt{\frac{0.5}{0.08}} = 2.5 \quad (50)$$

Physical Interpretation:

Light holes have longer wavelengths (smaller k) for the same energy because they have smaller effective mass. This means they are more "spread out" in real space and respond more readily to external forces, giving them higher mobility.

Answer: $k_{hh} = 0.115 \text{\AA}^{-1}$, $k_{lh} = 0.0458 \text{\AA}^{-1}$; wavelength ratio $\lambda_{lh}/\lambda_{hh} = 2.5$

Exercise 3.3: Negative Effective Mass at Band Maximum

Problem: Consider a valence band with dispersion relation $E(k) = E_v - Ak^2$, where $A = 2.0 \text{ eV} \cdot \text{\AA}^2$ and $E_v = 0 \text{ eV}$ (taking the valence band maximum as the reference). Calculate:

- (a) The effective mass of electrons at the band maximum
- (b) Explain why we describe these states as "holes" with positive effective mass

Solution**Step 1: Calculate the second derivative**

Given: $E(k) = E_v - Ak^2 = -Ak^2$ (with $E_v = 0$)

First derivative:

$$\frac{dE}{dk} = -2Ak \quad (51)$$

Second derivative:

$$\frac{d^2E}{dk^2} = -2A \quad (52)$$

Step 2: Calculate the effective mass for electrons

Using the definition:

$$\frac{1}{m_e^*} = \frac{1}{\hbar^2} \frac{d^2E}{dk^2} = \frac{-2A}{\hbar^2} \quad (53)$$

Therefore:

$$m_e^* = -\frac{\hbar^2}{2A} \quad (54)$$

Step 3: Calculate the numerical value

Using $\hbar^2/(2m_0) = 3.81 \text{ eV} \cdot \text{\AA}^2$:

$$m_e^* = -\frac{2m_0 \times 3.81 \text{ eV} \cdot \text{\AA}^2}{2 \times 2.0 \text{ eV} \cdot \text{\AA}^2} = -\frac{3.81m_0}{2.0} = -1.905m_0 \quad (55)$$

Answer to part (a): The electron effective mass is $m_e^* = -1.905m_0$ (negative!)

Step 4: Explain the hole concept

At the valence band maximum, the curvature is negative ($d^2E/dk^2 < 0$), meaning electrons at this band edge have *negative* effective mass. When an electric field is applied, these electrons would accelerate opposite to expectation.

Instead of tracking these peculiar electrons, we describe the absence of an electron (a hole) as a positive charge carrier. The hole effective mass is:

$$m_h^* = -m_e^* = -(-1.905m_0) = +1.905m_0 \quad (56)$$

Answer to part (b):

Holes are introduced as a convenient way to describe the collective behavior of many electrons near a filled band. Instead of tracking complicated dynamics of many electrons with negative effective mass, we track the missing electrons (holes) which behave as particles with:

- Positive charge ($+e$)
- Positive effective mass $m_h^* = +1.905m_0$
- Normal response to electric fields

This greatly simplifies semiconductor physics and device operation analysis.

Answer:

- (a) Electron effective mass: $m_e^* = -1.905m_0$
- (b) Holes with positive mass $m_h^* = +1.905m_0$ describe electron absences, simplifying physics since they respond normally to fields

Summary

These exercises cover fundamental concepts in quantum mechanics and solid state physics:

- **Hydrogen Atom:** Energy quantization, photon emission/absorption, ionization from different energy levels
- **Kronig-Penney Model:** Band structure, Brillouin zones, forbidden energy gaps, free electron limit
- **Effective Mass:** Parabolic band approximation, heavy vs. light holes, negative effective mass and the hole concept

These problems provide a foundation for understanding quantum mechanics in atoms and the electronic properties of crystalline solids. The step-by-step solutions are designed to help students develop intuition for these important physical concepts.